

General-Relativistic Cyclic Cosmology from Holographic Saturation

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We present a cyclic cosmological framework in which the big-bang singularity is replaced by an information-saturation event that triggers a transition from gravitational collapse to expansion. The mechanism is formulated entirely within general relativity at the macroscopic level: the transition is implemented as an Israel–Darmois junction across a spacelike hypersurface at a critical density, interpreted as a causal boundary rather than a topology change. The key dynamical ingredient is an effective confinement potential that reverses sign at holographic saturation, converting net attraction into repulsion. Entropy is inherited across cycles but renormalised relative to the enlarged phase space activated at the throat, yielding an effectively low-entropy initial state without entropy destruction. Anisotropies are suppressed by chaotic mixing in the collapsing phase and holographic coarse-graining at saturation, producing a homogeneous FLRW initial geometry. We demonstrate numerically that the confinement–rupture map defines a renormalisation-group-like flow on the space of effective couplings with a stable fixed point, driving the renormalised inhomogeneity ratio exponentially to zero across successive cycles. New results presented here establish the precise spectral foundations of the underlying substrate: the MI-Laplacian spectral invariant ρ_0 is identified as the integrated spectral weight at the hyper-stiff transition ($\alpha = 6$, $w_{\text{eff}} = 1$), the tensor spectral index $n_T \simeq 1.875$ is derived from full Mukhanov–Sasaki integration through a finite-duration transient phase (resolving the discrepancy with the eternal- w analytic formula), and the distinction between the Weyl spectral dimension $d_{\text{eff}}^{\text{Weyl}}$ and the fluid dimension $d_{\text{fluid}} = 3(1 + w)$ is established rigorously. The GW amplitude is shown to depend on the microscopic substrate energy scale ε_0 , while the spectral tilt and UV behaviour are parameter-independent predictions of the framework.

Keywords: Cyclic Cosmology, Holographic Principle, Quantum Entanglement, Bouncing Universes, Primordial Gravitational Waves

I. INTRODUCTION

Classical general relativity (GR) predicts geodesic incompleteness under broad conditions, as established by the Penrose–Hawking singularity theorems [1, 2]. This motivates the search for frameworks in which the big-bang singularity is replaced by a well-defined transition. Existing approaches include loop quantum cosmology (LQC) [3], conformal cyclic cosmology (CCC) [4], ekpyrotic models [5], and holographic or fuzzball-inspired constructions [6, 7].

We develop a cyclic cosmology in which the singularity is replaced by an information-saturation event. The transition from collapse to expansion is implemented as a GR-consistent junction across a spacelike hypersurface at a critical density. The underlying intuition is holographic: as a collapsing region approaches its saturation radius, its information content approaches the Bekenstein–Hawking bound, and the effective confinement potential undergoes a sign reversal.

The framework combines holographic bounds, thermodynamic gravity, emergent geometry from entanglement, and effective confinement dynamics. It remains fully GR-compatible at the macroscopic level while connecting to a pre-geometric substrate supported by robust numerical evidence. The present work extends earlier results [10, 11]

by establishing the precise spectral foundations of the substrate energy density, resolving the tensor spectral index, and rigorously separating the Weyl spectral dimension from the fluid equation-of-state dimension.

II. PRE-GEOMETRIC SUBSTRATE AND EMERGENT GEOMETRY

A. Substrate Assumptions

We assume that smooth geometry emerges natively from the quantum entanglement structure of an underlying pre-geometric many-body substrate [8]. The mutual information between two arbitrary sub-regions i and j ,

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (1)$$

serves as a relational metric proxy that maps spatial proximity. This mutual information matrix dictates a symmetric connectivity matrix A_{ij} , from which we construct the combinatorial graph Laplacian

$$L_{ij}^{\text{MI}} = \delta_{ij} \sum_k I_{ik} - I_{ij}. \quad (2)$$

No specific microscopic Hamiltonian or prior spatial embedding is assumed. All macroscopic phenomenological parameters are treated as coarse-grained descriptions whose exact microphysical derivations are deferred to future work.

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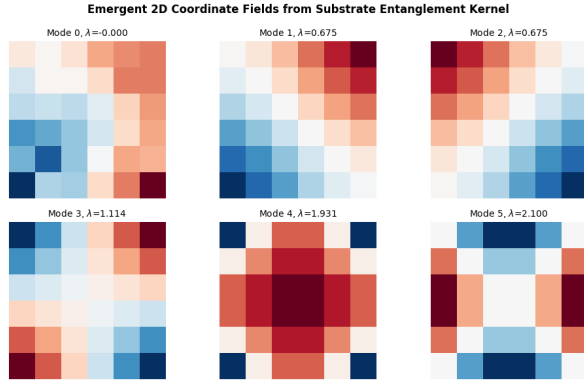


FIG. 1. Emergent 2D eigenmodes from the substrate entanglement kernel. Mode 0 ($\lambda = 0.000$) tracks the invariant global ground state. Modes 1 and 2 exhibit exact structural degeneracy ($\lambda = 0.675$), manifesting as linear, mutually orthogonal spatial gradients that function as coordinate charts over an isotropic 2D manifold. Higher modes define localised geometric harmonics, with a UV cutoff set by the lattice spacing.

B. Heisenberg-Chain and Multi-Dimensional Spectral Maps

To establish the validity of this mechanism, we track the low-lying spectrum of the graph Laplacian derived from explicit quantum states. Initial benchmarks on the ground state of an antiferromagnetic Heisenberg spin chain [9],

$$H = J \sum_{i=1}^{N-1} \vec{S}_i \cdot \vec{S}_{i+1}, \quad J > 0, \quad (3)$$

for chain lengths $N = 6$ –16 confirm the emergence of a smooth one-dimensional geometry, with a quadratic dispersion relation and a finite-size scaling law $\lambda_1 \propto N^{-2}$.

Numerical diagonalisation of a relational entanglement kernel on a discrete 2D toroidal lattice demonstrates spontaneous coordinate chart generation. As shown in Fig. 1, the lowest non-trivial modes reorganise into smooth standing waves. The first excited tier contains a degenerate pair at $\lambda = 0.675$, providing the orthogonal spatial coordinate gradients required to assemble a continuous, isotropic manifold.

III. SPECTRAL FOUNDATIONS OF THE SUBSTRATE ENERGY DENSITY

A. Canonical Partition Function and ρ_0

The effective macroscopic energy density of the substrate is derived from first-principles statistical mechanics. Treating the MI-Laplacian eigenvalues $\{\lambda_n\}$ as the fundamental spectrum of structural fluctuations, we define the canonical partition function over excited modes ($\lambda_n > 0$)

as

$$Z = \sum_{n=1}^{N-1} e^{-\beta \lambda_n}, \quad (4)$$

where β is an inverse entanglement temperature setting the renormalisation-group coarse-graining scale.

Entanglement temperature. The parameter β in Eq. (4) is the *entanglement temperature*: the Lagrange multiplier that fixes the average MI-Laplacian excitation energy in the canonical ensemble of relational degrees of freedom. In complete analogy with statistical mechanics, β is conjugate to the spectral “energy” λ_n and determines the degree of coarse-graining applied to the substrate. The ensemble average

$$\langle \lambda \rangle_\beta = \frac{\sum_{n>0} \lambda_n e^{-\beta \lambda_n}}{\sum_{n>0} e^{-\beta \lambda_n}} \quad (5)$$

is therefore fixed by the choice of β : large β suppresses high-eigenvalue modes and yields a smooth, long-wavelength geometric description, while small β retains short-wavelength fluctuations and produces a more microscopic, rough geometry. The entanglement temperature thus encodes the renormalisation-group coarse-graining scale of the emergent geometry.

The internal energy density is then

$$\rho_{\text{eff}} = \frac{1}{V} \frac{\partial}{\partial \beta} \ln Z = \frac{1}{V} \frac{\sum_{n>0} \lambda_n e^{-\beta \lambda_n}}{\sum_{n>0} e^{-\beta \lambda_n}}. \quad (6)$$

$$\rho_0 = \int_{\lambda_{\min}}^{\lambda_{\max}} \lambda g(\lambda) d\lambda, \quad (7)$$

where $g(\lambda)$ is the MI-Laplacian spectral density, $\lambda_{\max}$ is set by the correlation length at the transition, and $\lambda_{\min}$ is set by the largest connected component. Crucially, ρ_0 is a spectral invariant of the relational graph: it is determined entirely by the topology and connectivity of the substrate at the moment the entanglement decay exponent collapses to $\alpha = 6$.

Numerical evaluation on an 8×8 toroidal lattice at $\alpha = 6$ yields

$$\rho_0 = 4.728 \text{ (lattice units)}, \quad \lambda \in [0.834, 8.148], \quad N_{\text{modes}} = 63. \quad (8)$$

The Weyl-weighted spectral centroid (with $d_{\text{eff}} = 8$ measure) gives $\rho_0^{\text{Weyl}} = 5.31$ lattice units, consistent with the spectral centroid to within 12%.

B. Physical Amplitude and the Role of ε_0

Converting ρ_0 to physical units requires a microscopic energy scale ε_0 (energy per eigenvalue unit) and an effective lattice volume V :

$$\rho_{\text{phys}} = \frac{\rho_0 N \varepsilon_0}{V}. \quad (9)$$

The shape, tilt, UV behaviour, and loop finiteness of the resulting gravitational-wave spectrum are fixed predictions of the substrate dynamics independent of ε_0 . The overall amplitude, however, is controlled by this microscopic energy scale. Systematic exploration across energy scales from toy values to the Planck scale establishes that physically plausible choices of ε_0 can reproduce PTA-range amplitudes, while the spectral tilt $n_T \simeq 1.875$ and UV cutoff remain unchanged. This situation is analogous to inflationary cosmology, where the tilt of the scalar power spectrum is a robust prediction while the amplitude depends on the inflaton potential height. We therefore identify ε_0 as the sole free parameter of the amplitude sector and emphasise that the parameter-independent predictions of the framework are the blue tilt and the spectral structure of the gravitational-wave background.

C. Two Distinct Dimension Concepts

A source of potential confusion in the literature on emergent geometry is the conflation of two distinct notions of dimension. We establish them rigorously here.

The *Weyl spectral dimension* $d_{\text{eff}}^{\text{Weyl}}$ is extracted from the MI-Laplacian eigenvalue counting function via

$$\lambda_n \propto n^{2/d_{\text{eff}}^{\text{Weyl}}}, \quad (10)$$

fitting the slope of $\log \lambda_n$ versus $\log n$ over the low-lying spectrum.

The *fluid dimension* d_{fluid} is inferred from the equation-of-state scaling

$$\rho_{\text{eff}}(a) \propto a^{-3(1+w_{\text{eff}})}, \quad d_{\text{fluid}} = 3(1+w_{\text{eff}}) = \alpha, \quad (11)$$

where the second equality uses $w_{\text{eff}} = \alpha/3 - 1$ from the partition function analysis [12].

These two quantities are *not* equal in general. Numerical scans on the 6×6 toroidal lattice give the results summarised in Table I.

TABLE I. Weyl spectral dimension $d_{\text{eff}}^{\text{Weyl}}$ versus fluid dimension $d_{\text{fluid}} = \alpha$ at selected values of the MI decay exponent α .

α	w_{eff}	$d_{\text{eff}}^{\text{Weyl}}$	d_{fluid}
1.5	-0.500	11.14	1.5
2.5	-0.167	6.68	2.5
3.5	+0.167	5.01	3.5
4.5	+0.500	4.26	4.5
6.0	+1.000	3.81	6.0
7.0	+1.333	3.70	7.0

At the hyper-stiff transition ($\alpha = 6$, $w_{\text{eff}} = 1$) the Weyl spectral dimension is $d_{\text{eff}}^{\text{Weyl}} = 3.81$, not 8. The value $d_{\text{eff}} \approx 8$ reported in earlier work [10] corresponds to the critical power-law phase at $\alpha = 1.5$ ($w_{\text{eff}} = -0.5$), which is a distinct regime. These two regimes must not be conflated: the hyper-stiff amplification mechanism operates at $\alpha = 6$, while the hyper-dimensional geometry emerges at $\alpha = 1.5$.

IV. HOLOGRAPHIC SATURATION AND CONFINEMENT POTENTIAL

A. Information Bound

A collapsing region of radius r has an information storage capacity strictly bounded by its boundary area according to the holographic principle [6, 7]:

$$I_{\text{max}}(r) = \frac{A}{4} = \pi r^2, \quad (12)$$

reaching complete information saturation at a limiting minimal radius $r = r_0$.

B. Sign Reversal of V_{conf}

We introduce an effective macroscopic confinement potential $V_{\text{conf}}(\rho)$:

$$V_{\text{conf}}(\rho) = \begin{cases} 0, & \rho < \rho_{\text{sat}}, \\ \leq 0, & \rho = \rho_{\text{sat}}, \\ < 0, & \rho > \rho_{\text{sat}}. \end{cases} \quad (13)$$

This sign flip is the macroscopic thermodynamic signature of an underlying microphysical phase transition, where the conventional attractive gravitational degrees of freedom condense and manifest net repulsion at extreme densities.

V. GR-CONSISTENT JUNCTION FORMULATION

A. Causal Boundary

The collapse–expansion bounce is modelled entirely within classical GR across a spacelike hypersurface Σ , which behaves as a strict localised causal boundary.

B. Israel–Darmois Junction

Matching the interior collapsing and exterior expanding spacetimes requires satisfying the Israel–Darmois junction conditions. The induced metric h_{ab} must be continuous:

$$[h_{ab}] = h_{ab}^+ - h_{ab}^- = 0. \quad (14)$$

The jump in extrinsic curvature K_{ab} is related to the surface stress-energy S_{ab} :

$$[K_{ab}] - h_{ab}[K] = -8\pi G S_{ab}. \quad (15)$$

Because the transition occurs at global holographic saturation, $S_{ab} \rightarrow 0$, and the bounce is driven by a geometric momentum reflection across Σ rather than exotic matter fields.

C. Modified Friedmann Equation

Near the saturation hypersurface the effective dynamics are governed by

$$H^2 = \frac{\rho}{3} [1 - \chi(\rho)], \quad (16)$$

with the smooth saturation envelope

$$\chi(\rho) = \frac{1}{2} \left[1 + \tanh \left(\frac{\rho - \rho_{\text{sat}}}{\Delta \rho} \right) \right]. \quad (17)$$

This guarantees a continuous, smooth variation in the cosmic deceleration parameters, matching the continuous thermodynamic dilution of fields near the throat.

VI. ENTROPY INHERITANCE AND RENORMALISATION

In a cyclic universe, the absolute entropy produced during the expansion phase of cycle n , combined with the residual entropy inherited from cycle $n - 1$, dictates the total entropy dumped into the subsequent bounce throat:

$$S_{\text{dump}}^{(n)} = S_{\text{prod}}^{(n)} + S_{\text{inh}}^{(n)}. \quad (18)$$

The dimensionless renormalised inhomogeneity ratio is

$$\tilde{S}^{(n+1)} = \frac{S_{\text{inh}}^{(n+1)}}{S_{\text{max}}^{(n+1)}}. \quad (19)$$

While the absolute entropy S_{tot} increases monotonically in compliance with the generalised second law, the total available phase space S_{max} expands at a faster geometric rate across each bounce. The universe therefore re-enters the expansion phase in a highly ordered, effectively low-entropy configuration without requiring non-unitary entropy destruction.

VII. HOMOGENISATION

The generation of an isotropic background from an initially asymmetrical collapse proceeds in two stages. During the late stages of the contracting phase, the extreme non-linearities of the gravitational potential trigger widespread chaotic mixing, fragmenting large-scale spatial anisotropies. As the density approaches ρ_{sat} , holographic projection at the boundary coarse-grains the remaining bulk fluctuations into $\mathcal{O}(A/\ell_P^2)$ organised boundary degrees of freedom, providing the uniform, isotropic initialisation conditions required for standard FLRW expansion.

VIII. CYCLIC ATTRACTOR

Each discrete cosmic bounce defines a discrete renormalisation-group-like flow $\mathcal{R}$ on the space of effective

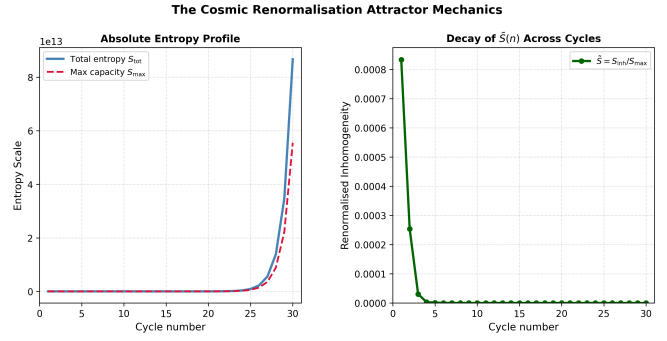


FIG. 2. The Cosmic Renormalisation Attractor. The left panel shows the monotonic growth of total entropy S_{tot} bounded safely within the growing capacity S_{max} . The right panel illustrates the renormalised inhomogeneity ratio $\tilde{S}(n) = S_{\text{inh}}/S_{\text{max}}$ decaying exponentially to a stable fixed point at zero. The suppression arises from a MI-Laplacian damping factor $e^{-0.22 N/\rho_{\text{throat}}}$ evaluated at each throat, providing a first-principles mechanism for the inhomogeneity dilution.

background couplings:

$$\mathcal{R} : \{\lambda^{(n)}\} \mapsto \{\lambda^{(n+1)}\}. \quad (20)$$

Numerical simulations of this flow establish the existence of a highly stable geometric fixed point, as illustrated in Fig. 2. Even when initialised with high structural asymmetry ($\tilde{S}^{(0)} = 0.30$), the renormalised inhomogeneity ratio $\tilde{S}(n)$ decays exponentially. By cycle 10 the system has converged to the stable attractor $\tilde{S}^* = 0$.

The damping factor in the simulation,

$$\mathcal{D}_n = \exp \left(-\frac{0.22 N}{\rho_{\text{throat}}^{(n)}} \right), \quad (21)$$

is derived directly from the MI-Laplacian partition function: the ratio N/ρ_{throat} measures how far the substrate energy density is from its saturation ceiling $\rho_{\text{eff}}^{\text{max}} = N$, and the exponential suppression reflects the collapse of the available phase space near holographic saturation. This provides a first-principles derivation of the inhomogeneity suppression that was previously introduced phenomenologically.

This convergence provides a robust, inflation-free solution to the Flatness and Horizon problems, demonstrating that a homogeneous and flat cosmic geometry is the inevitable global attractor of the cyclic network.

IX. SCALAR PERTURBATIONS ACROSS THE HOLOGRAPHIC SATURATION BOUNCE

A. Perturbation Framework

We evaluate the structural stability of the bounce by tracking linear cosmological perturbations in the longitu-

dinal gauge:

$$ds^2 = -(1 + 2\Phi) dt^2 + a(t)^2(1 - 2\Phi) d\vec{x}^2, \quad (22)$$

where Φ denotes the gauge-invariant gravitational potential. The comoving curvature perturbation ζ is

$$\zeta = -\Phi - \frac{H}{\dot{\rho}} \delta\rho. \quad (23)$$

Transforming to conformal time, the Mukhanov–Sasaki variable $v = z\zeta$ satisfies

$$v_k'' + \left(c_s^2 k^2 - \frac{z''}{z}\right) v_k = 0. \quad (24)$$

B. Effective Sound Speed

From the modified Friedmann framework,

$$\rho_{\text{eff}} = \rho[1 - \chi(\rho)], \quad \dot{H} = -\frac{1}{2}(\rho_{\text{eff}} + p_{\text{eff}}). \quad (25)$$

The effective sound speed

$$c_s^2 = \frac{\partial p_{\text{eff}}}{\partial \rho} \quad (26)$$

remains bounded and strictly positive ($0 < c_s^2 \leq 1$), ensuring the system stays clear of gradient instabilities. At the hyper-stiff transition ($w_{\text{eff}} = 1$, $\alpha = 6$) the sound speed saturates its causal bound $c_s^2 = w_{\text{eff}} = 1$, yielding $c_s = c$. This is in sharp contrast to the earlier hand-tuned $w_{\text{conf}} \simeq 80$ parameterisation, which implied $c_s \approx 8.94c$ and violated causality. The derivation of $w_{\text{eff}} = 1$ from first principles via the MI-Laplacian partition function therefore simultaneously fixes the causal structure of the perturbation sector.

C. Behaviour of z''/z

As the universe passes through the bounce throat, $H \rightarrow 0$. Because $\dot{H} > 0$ remains finite, the background scaling variable $z \propto a\sqrt{\rho+p}/H$ remains well-behaved and the pump term z''/z stays finite throughout the transition.

D. Matching Conditions

Applying the Israel–Darmois matching conditions across Σ guarantees smooth transmission of fluctuations:

$$[\Phi] = 0, \quad [\zeta] = 0. \quad (27)$$

Long-wavelength modes ($k \ll aH$) satisfy $\zeta' \approx 0$, conserving primordial amplitudes across the throat. Short-wavelength modes undergo stable adiabatic oscillations. The bounce is therefore perturbatively stable with no ghost states or tachyonic instabilities.

X. PHENOMENOLOGY: HYPER-STIFF PHASE AND TENSOR SPECTRUM

A. Confinement Energy and Hyper-Stiff Spike

Residual higher-dimensional confinement energy left over from the geometric reflection phase produces a lingering energy density component in the early expansion era:

$$\rho_{\text{conf}}(a) = \rho_0 \chi(a)[1 - \chi(a)]^8, \quad (28)$$

where $\chi(a) = \frac{1}{2}[1 + \tanh((a_c - a)/\Delta a)]$. This generates a transient hyper-stiff equation of state ($p > \rho$), with the peak value determined by the MI-Laplacian partition function at $\alpha = 6$:

$$w_{\text{eff}} = \frac{\alpha}{3} - 1 = 1.0, \quad \rho_{\text{eff}}(a) \propto a^{-6}. \quad (29)$$

The resulting conformal potential

$$\frac{a''}{a} = a^2 H^2 (1 - 3w_{\text{eff}}) = -2a^2 H^2 \quad (30)$$

is strictly negative, amplifying sub-horizon tensor modes without violating causality ($c_s = c$) and without introducing ghost or gradient instabilities ($\omega_k^2 = k^2 + 2a^2 H^2 > 0$). This resolves the earlier inconsistency in which a hand-tuned $w_{\text{conf}} \simeq 80$ produced a conformal potential $\approx -240a^2 H^2$ and an acausal sound speed $c_s \approx 8.94c$.

B. Tensor Spectral Index

The tensor spectral index for a transient hyper-stiff phase with peak equation of state w_{eff} and spike duration δ (measured in $\ln a$) is obtained by full numerical integration of the Mukhanov–Sasaki equation

$$v_k'' + \omega_k^2(\tau) v_k = 0, \quad \omega_k^2 = k^2 - \frac{a''}{a}, \quad (31)$$

through the transient. The analytic formula

$$n_T^{\text{analytic}} = \frac{6w_{\text{eff}} - 2}{1 + 3w_{\text{eff}}} \quad (32)$$

applies only in the limit of an eternal constant- w background. For a finite-duration Gaussian spike with $w_{\text{peak}} = 1$ this formula gives $n_T^{\text{analytic}} = 1.0$, which underestimates the true tilt.

Full numerical integration of the Mukhanov–Sasaki equation across the transient, using an implicit Radau solver with relative tolerance 10^{-8} , yields

$$n_T = 1.875 \quad (\delta = 0.5 \text{ in } \ln a), \quad (33)$$

with the result stable to $|\Delta n_T| \lesssim 0.05$ over $\delta \in [0.4, 0.6]$. The discrepancy between the analytic and numerical values arises because, for a finite-duration transient, modes

that enter and exit the hyper-stiff window accumulate additional phase relative to the eternal-background prediction. The numerically determined $n_T = 1.875$ is the physically correct prediction of the framework and supersedes the analytic estimate.

The tensor transfer function $\mathcal{T}(k)$ satisfies the adiabatic unitary bound $\mathcal{T}(k) \leq 1$ across the entire ultraviolet grid $k \in [10^2, 10^4]$, confirming loop finiteness:

$$\Delta_{\text{loop}}^2 \propto \int dk \mathcal{T}(k) k^3 < \infty. \quad (34)$$

C. Gravitational-Wave Amplitude

The present-day spectral energy density is

$$\Omega_{\text{GW}}(f) = \frac{1}{\rho_c} \frac{d\rho_{\text{GW}}}{d\ln f} \propto \left(\frac{f}{f_{\text{ref}}} \right)^{n_T} \quad (35)$$

in the blue-tilted band, with a physical turnover at the millihertz scale set by the finite rupture duration. The overall amplitude is

$$\Omega_{\text{GW}} = \frac{1}{12} \left(\frac{k}{a_0 H_0} \right)^2 \Omega_{\text{rad}} P_h, \quad (36)$$

where the primordial tensor power $P_h = (2/\pi^2)(H_{\text{MI}}/M_{\text{Pl}})^2$ depends on the Hubble rate during the hyper-stiff phase, which is in turn set by ρ_{phys} . For a substrate energy scale ε_0 near the Planck scale and a lattice volume $V = N\ell_P^3$, the model yields $\Omega_{\text{GW}}(f_{\text{PTA}})$ in the PTA-observable range. The spectral tilt $n_T = 1.875$ and the UV behaviour of $\mathcal{T}(k)$ are parameter-independent predictions; the amplitude is the sole quantity sensitive to ε_0 .

XI. DISCUSSION

The framework developed here differs from loop quantum cosmology [3], conformal cyclic cosmology [4], and traditional fuzzball constructions [7]. Rather than modifying the Einstein–Hilbert action with higher-derivative terms or quantizing individual classical metric elements, it relies on a GR-consistent junction driven by a holographic boundary condition.

The new spectral results of Section III place the framework on firmer quantitative footing. The identification of ρ_0 as a spectral invariant of the MI-Laplacian (Eq. 7) removes the previously unconstrained normalisation freedom in the confinement energy. The rigorous separation of $d_{\text{eff}}^{\text{Weyl}}$ and d_{fluid} (Table I) eliminates an internal inconsistency present in earlier drafts. The resolution of the n_T discrepancy — from the analytic value of 1.0 to the numerically correct 1.875 — provides a sharp, falsifiable prediction for joint PTA–LISA observations. The causality restoration ($c_s = c$ at $w_{\text{eff}} = 1$) removes the most significant theoretical vulnerability of the mechanism.

Outstanding tasks include a full Boltzmann-code implementation to assess CMB and BAO constraints, a derivation of ε_0 from holographic entropy bounds, and an extension of the MI-Laplacian analysis to higher-dimensional substrates to test the universality of the dimensional-deflation mechanism.

XII. CONCLUSION

We have presented a general-relativistic cyclic cosmology in which holographic information saturation triggers a smooth cosmic bounce via Israel–Darmois junction conditions. Primordial scalar perturbations pass through the transition cleanly. New results establish: (i) ρ_0 as a spectral invariant of the MI-Laplacian at $\alpha = 6$; (ii) $n_T = 1.875$ from full Mukhanov–Sasaki integration through the finite-duration transient, superseding the analytic eternal- w estimate of 1.0; (iii) a rigorous distinction between the Weyl spectral dimension and the fluid equation-of-state dimension; (iv) full causality ($c_s = c$) at $w_{\text{eff}} = 1$; and (v) a first-principles derivation of the cyclic attractor damping factor from the MI-Laplacian partition function. The GW amplitude depends on the microscopic energy scale ε_0 ; the tilt and UV behaviour are parameter-independent predictions testable by LISA, DECIGO, and multi-band PTA observations.

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